Pipelined Performance Example

- A program running on a 5-stage pipelined processor has been profiled and consists of:
  - ✓ 25% loads
  - ✓ 10% stores
  - ✓ 11% conditional branches
  - ✓ 2% jumps (unconditional branches)
  - ✓ 52% R-type (arithmetic or logic operations)
- Assume:
  - ✓ Early branch resolution in DECODE
  - ✓ No branch prediction; flush of incorrectly-fetched instructions
  - ✓ 40% of loads used by next instruction
  - ✓ 25% of branches mispredicted
  - ✓ All jumps flush the next instruction
- What is the **average clock-per-instruction (CPI)?**

Solution

- Load/Branch CPI = 1 when no stall/flush, 2 when stall/flush. Thus:
  - ✓ $CPI_{load} = 1(0.6) + 2(0.4) = 1.4$ *Average CPI for load*
  - ✓ $CPI_{branch} = 1(0.75) + 2(0.25) = 1.25$ *Average CPI for branch*
- Thus,

Average CPI =	(0.25)(1.4) +	*load*
	(0.1)(1) +	*store*
	(0.11)(1.25) +	*beq*
	(0.02)(2) +	*jump*
	(0.52)(1)	*R-type*
	= **1.15**	

Execution Time

- For a program with 100 billion instructions executing on this pipelined processor, assuming a clock cycle time of Tc = 550 ps,
- What is the execution time of the program?

Solution

We know the average CPI = 1.15.

We can therefore apply the performance equation:

Execution Time $= (\#\text{ instructions}) \times \text{Avg. CPI} \times T_c$

$= (100 \times 10^9)(1.15)(550 \times 10^{-12})$

$=$ 63 seconds

Single-Cycle Processor

- With a single-cycle processor, what would be the maximum clock period T_c in order to execute the same 100-billion instruction program with an equal or lower execution time?

Solution

For a single-cycle processor

- Avg. CPI = 1
- We expect that the clock cycle time will be >550 ps

Let us solve the performance equation with respect to the clock cycle time Tc:

$$63 \text{ seconds} = (\#\text{ instructions}) \times \text{Avg. CPI} \times T_c$$
$$= (100 \times 10^9)(1)(T_c)$$

Thus

$T_c <= 630$ psec